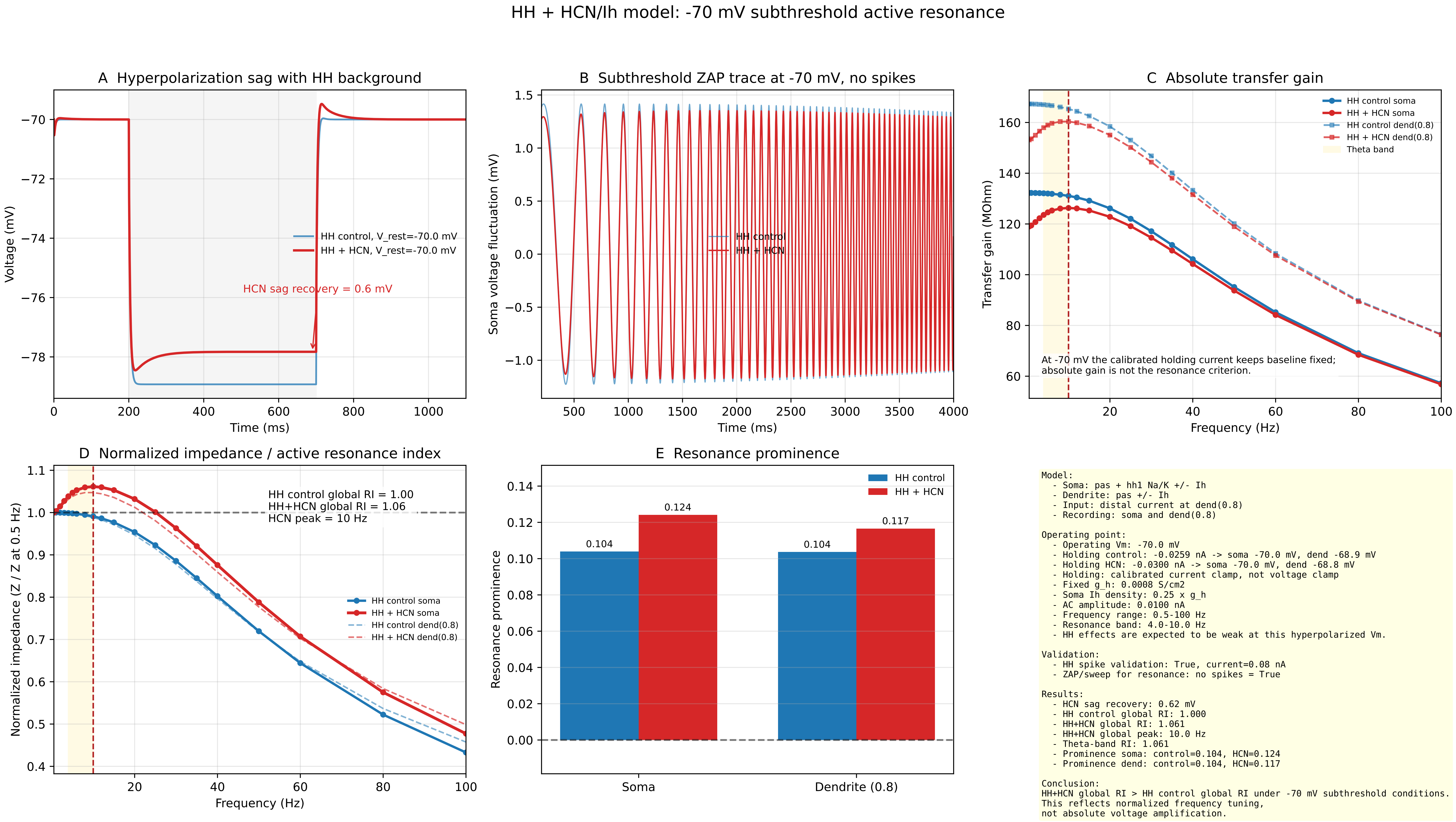
HH + HCN/Ih model: -70 mV subthreshold active resonance
A Hyperpolarization sag with HH background
Voltage (mV)
Time (ms)
HH control, V_rest=-70.0 mV
HH + HCN, V_rest=-70.0 mV
HCN sag recovery = 0.6 mV
B Subthreshold ZAP trace at -70 mV, no spikes
Soma voltage fluctuation (mV)
Time (ms)
HH control
HH + HCN
C Absolute transfer gain
Transfer gain (MOhm)
Frequency (Hz)
HH control soma
HH + HCN soma
HH control dend(0.8)
HH + HCN dend(0.8)
Theta band
At -70 mV the calibrated holding current keeps baseline fixed;
absolute gain is not the resonance criterion.
D Normalized impedance / active resonance index
Normalized impedance (Z / Z at 0.5 Hz)
Frequency (Hz)
HH control global RI = 1.00
HH+HCN global RI = 1.06
HCN peak = 10 Hz
HH control soma
HH + HCN soma
HH control dend(0.8)
HH + HCN dend(0.8)
E Resonance prominence
Resonance prominence
HH control
HH + HCN
0.104
0.124
0.104
0.117
Soma
Dendrite (0.8)
Model:
- Soma: pas + hh1 Na/K +/- Ih
- Dendrite: pas +/- Ih
- Input: distal current at dend(0.8)
- Recording: soma and dend(0.8)
Operating point:
- Operating Vm: -70.0 mV
- Holding control: -0.0259 nA -> soma -70.0 mV, dend -68.9 mV
- Holding HCN: -0.0300 nA -> soma -70.0 mV, dend -68.8 mV
- Holding: calibrated current clamp, not voltage clamp
- Fixed g_h: 0.0008 S/cm2
- Soma Ih density: 0.25 x g_h
- AC amplitude: 0.0100 nA
- Frequency range: 0.5-100 Hz
- Resonance band: 4.0-10.0 Hz
- HH effects are expected to be weak at this hyperpolarized Vm.
Validation:
- HH spike validation: True, current=0.08 nA
- ZAP/sweep for resonance: no spikes = True
Results:
- HCN sag recovery: 0.62 mV
- HH control global RI: 1.000
- HH+HCN global RI: 1.061
- HH+HCN global peak: 10.0 Hz
- Theta-band RI: 1.061
- Prominence soma: control=0.104, HCN=0.124
- Prominence dend: control=0.104, HCN=0.117
Conclusion:
HH+HCN global RI > HH control global RI under -70 mV subthreshold conditions.
This reflects normalized frequency tuning,
not absolute voltage amplification.